

tracert: This tool traces the path packets take from the source to the destination, listing each hop along the way. It identifies delays and routing paths.

Usage benefits	Example command
Pinpoints problematic network hops	<code>tracert google.com</code>
Assesses latency at each hop	<code>tracert -l 8.8.8.8</code>
Identifies routing paths	<code>tracert example.com</code>

Diagnosing DNS issues

nslookup: This tool is used to query the Domain Name System (DNS) and retrieve details like IP addresses associated with domain names.

Usage benefits	Example command
Verifies DNS resolution	<code>nslookup google.com</code>
Retrieves IP addresses for domains	<code>nslookup example.com</code>
Queries specific DNS servers	<code>nslookup example.com 8.8.8.8</code>

Dig (Domain Information Groper): This offers advanced DNS diagnostics, providing detailed query and response information.

Usage benefits	Example command
Offers robust DNS diagnostics	<code>dig google.com</code>
Performs reverse DNS lookups	<code>dig -x 8.8.8.8</code>
Provides detailed query outputs	<code>dig +short example.com</code>

Route and gateway checking

ip route: The `ip route` command is used for displaying and managing the routing table. It helps ensure proper routing configurations.

Usage benefits	Example command
Displays routing table	<code>ip route show</code>
Adds a static route	<code>ip route add 192.168.1.0/24 via 10.0.0.1</code>
Deletes a route	<code>ip route del 192.168.1.0/24</code>

These tools provide a solid foundation for troubleshooting and maintaining basic network connectivity. They ensure administrators can address issues related to reachability, DNS resolution, and routing effectively.

Network interface configuration and monitoring

Network interface configuration and monitoring are critical tasks for ensuring efficient network operation. Administrators use these tools to manage interfaces, monitor traffic, and troubleshoot

performance issues in real-time. Effective interface management ensures connectivity stability, while real-time monitoring helps identify bottlenecks and unusual traffic patterns.

Key questions administrators face daily with respect to network interface configuration are:

- Are all network interfaces configured and functioning correctly?
- Is there abnormal traffic activity affecting the network's performance?

Interface management

ip: The `ip` command is a versatile tool for managing IP addresses, routes, and network interfaces. It replaces the older `ifconfig` command and offers extended functionality for modern networking.

Usage benefits	Example command
Configures network interfaces	<code>ip addr add 192.168.1.10/24 dev eth0</code>
Displays interface details	<code>ip addr show</code>
Enables or disables interfaces	<code>ip link set dev eth0 up/down</code>

nmcli: nmcli is a command-line tool for controlling the network manager. It allows administrators to manage network connections without a graphical user interface, making it ideal for remote systems.

Usage benefits	Example command
Lists active network connections	<code>nmcli connection show</code>
Modifies a connection	<code>nmcli connection modify conn-name ...</code>
Activates or deactivates connections	<code>nmcli connection up/down conn-name</code>

Real-time monitoring

iftop: iftop provides a real-time view of bandwidth usage per connection, displayed in a top-like interface. It is invaluable for identifying bandwidth-heavy applications and devices.

Usage benefits	Example command
Monitors bandwidth usage	<code>sudo iftop</code>
Identifies heavy bandwidth users	<code>sudo iftop -i eth0</code>
Filters traffic by host	<code>sudo iftop -f 192.168.1.0/24</code>

nload: nload is a console-based tool for real-time traffic monitoring. It displays incoming and outgoing traffic statistics in an easy-to-read format.